A Dense Metrizable Core in Every Possible Zero-Dimensional Model

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Source Question

The source is Grzegorz Plebanek, *Musing on Kunen's compact L-space*, arXiv:2012.01849 [1]. Construction 3.3 produces two connected Corson compact L-spaces, according to whether the normal measure used at each successor stage has Maharam type ω_1 or ω. For the first version, Theorem 4.3 proves that $C(K)$ is isomorphic to no $C(L)$ with L zero-dimensional. The final paragraph on page 10 asks whether this remains true for the countable-type version:

> "We do not know if $C(K)$ satisfies the assertion of Theorem 4.3."

every n,

$$C(K) \ni f \to Tf(y_n),$$

is a continuous functional on $C(K)$ which is represented by a signed measure ν_n on K. Writing every ν_n as $\nu_n = \nu_n^+ - \nu_n^-$ (as a difference of two nonnegative measures), we get a countable family $\{\nu_n^+, \nu_n^- : n < \omega\}$ of nonnegative measures separating elements of $C(K)$. $\square$

One can check that if we let K to be connected L-space supporting a measure μ of countable type then $C(K)^*$ has a *weak** separable dual unit ball; in particular, Lemma 4.2 does not hold. We do not know if $C(K)$ satisfies the assertion of Theorem 4.3. Let us recall that Talagrand's construction mentioned in the introductory section gives, under CH, a tricky space K for which $C(K)^*$ is *weak** separable while the dual unit ball in $C(K)^*$ is not. Such an example was later constructed in the usual set theory, see [2].

References

The source explains why its last separating-measures lemma fails in the countable-type version: the dual unit ball of $C(K)^*$ is weak-star separable.

Result Type and Scope

This packet gives a **substantial partial reduction**. It does not decide whether a zero-dimensional model exists. It proves that every such model is metrizable on a dense open cozero set, and confines all possible nonmetrizability to a nowhere-dense zero-set boundary. The result applies to both measure-type versions of the source construction.

Proof Intuition

Two source theorems survive the change of Maharam type. First, an isomorphism $C(K) \cong C(L)$ forces L to have a π-base whose closures are continuous images of compact subspaces of K. Second, every zero-dimensional compact image of a closed subspace of this K is metrizable. Hence the π-base has metrizable closures.

Now use ccc globally. A maximal disjoint family from that π-base is countable, and maximality makes its union dense. The union is a countable topological sum of second-countable open spaces, hence metrizable. A small extra observation—an open subset of a compact metrizable closure is F_σ in the ambient compactum—makes the union cozero. This last step turns the topological localization into an exact $C(L)$-space reduction to a zero-set boundary.

General Localization Theorem

Theorem 1. *Let K and L be compact Hausdorff spaces. Assume that*

1. *K is ccc;*
2. *every zero-dimensional compact space that is a continuous image of a closed subspace of K is metrizable;*
3. *L is zero-dimensional and $C(K)$ is isomorphic to $C(L)$.*

Then L contains a dense open metrizable cozero subspace O. Consequently $F = L \setminus O$ is a nowhere-dense zero-set, $C_0(O)$ is separable, and restriction to F gives an exact sequence

$$0 \longrightarrow C_0(O) \longrightarrow C(L) \longrightarrow C(F) \longrightarrow 0.$$

Proof. Rosenthal's characterization says that ccc of a compactum S is equivalent to separability of every weakly compact subset of $C(S)$ [3]. This is an isomorphic property of the Banach space. Thus L is ccc.

Apply Plebanek's isomorphism theorem with the roles of K and L interchanged [2]. There is a π-base $\mathcal{V}$ of L such that, for every $V \in \mathcal{V}$, the compactum $\overline{V}$ is a continuous image of a compact subspace of K. A compact subspace is closed in K. Since L is zero-dimensional, so is its closed subspace $\overline{V}$. Hypothesis (2) therefore makes every $\overline{V}$ metrizable.

Choose, by Zorn's lemma, a maximal pairwise disjoint subfamily $\mathcal{W} \subseteq \mathcal{V}$. Since L is ccc, $\mathcal{W} = \{V_n : n \in I\}$ for some countable set I. Put

$$O = \bigcup_{n \in I} V_n.$$

The set O is dense. Indeed, if a nonempty open $U \subseteq L$ were disjoint from O, the π-base property would give $V \in \mathcal{V}$ with $V \subseteq U$; adjoining V would contradict maximality of $\mathcal{W}$.

Every V_n is second countable because it is an open subspace of the compact metrizable space $\overline{V_n}$. Within O, the sets V_n are pairwise disjoint clopen pieces. Hence O is their countable topological sum and is second countable. It is regular as a subspace of L, so the Urysohn metrization theorem makes O metrizable.

It remains to record the useful cozero refinement. The set V_n is open in the metric compactum $\overline{V_n}$, so it is F_σ there. Each closed term in such a representation is compact, hence closed in L. Thus every V_n, and therefore O, is an open F_σ subset of L. In a compact Hausdorff (hence normal) space every open F_σ set is a cozero set: if $O = \bigcup_m F_m$ with F_m closed, use Urysohn functions f_m that are 1 on F_m and 0 off O, and sum a uniformly convergent positive multiple of the f_m. Consequently $F = L \setminus O$ is a zero-set; density of O makes F nowhere dense.

Finally, a locally compact second-countable space has separable C_0, so $C_0(O)$ is separable. The restriction map $C(L) \to C(F)$ is onto by Tietze extension, and its kernel is canonically $C_0(O)$. This proves the exact sequence. □

Application to the Countable-Type Kunen Space

Corollary 1. *Let K be the connected compact L-space obtained from Construction 3.3 of [1] using the measures ν_2, so that its strictly positive normal measure has countable Maharam type. If L is zero-dimensional and $C(K) \cong C(L)$, then L has the decomposition and exact sequence asserted in Theorem 1.*

Proof. The strictly positive finite measure on K makes K ccc: a disjoint family of nonempty open sets is a disjoint family of positive-measure sets and hence is countable. Theorem 3.4 of [1], whose proof uses only the common monotone inverse-system construction and not the Maharam type, says exactly that every zero-dimensional compact continuous image of a closed subspace of K is metrizable. Theorem 1 applies. □

In particular, the source's separability conclusion for a hypothetical L can be sharpened to a dense *open metrizable cozero core*. The nonmetrizable part, if it exists, cannot meet any open set exclusively; it is forced into the zero-set boundary F.

Full-Upgrade Audit

Six focused attempts were made to remove the remaining boundary.

1. The source's separating-measures proof cannot be repaired: the countable-type construction deliberately has weak-star separable dual unit ball, so countably many measures may separate $C(K)$.
2. One cannot force L to be Corson through WLD invariance. A Corson compactum carrying a measure with nonseparable support need not make $C(K)$ WLD; the relevant property-M bridge is absent here.
3. Passing to $F = L \setminus O$ gives the exact sequence above, but an arbitrary isomorphism transports $C_0(O)$ only to a separable Banach subspace of $C(K)$, not to a topological ideal or a closed subset of K.
4. Even if the zero-set boundary admits an extension operator in a particular model, a complemented copy of $C(F)$ inside $C(K)$ does not contradict a known isomorphic invariant of the countable-type construction.
5. Replacing interval fibers by Cantor fibers stage-by-stage invokes Milutin isomorphisms only at individual metrizable stages. No coherent transfinite intertwining of the successor extension classes follows.
6. Quantitative Banach–Stone theory rules out isomorphisms of distortion strictly below 2, but the source asks about unrestricted distortion and the threshold is sharp in general.

These obstructions leave no credible bounded route to a full proof or counterexample in this pass.

Verification and Novelty Status

The proof uses only the three explicitly cited source ingredients and elementary compact-space facts. The ccc/maximal-family step, the F_σ-to-cozero step, and the $C_0(O)$ kernel identification were checked independently.

The run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2012.01849 and the core Kunen/countable-type/$C(K)$ terms. Bounded external searches used the exact final sentence and the phrases "dense open metrizable $C(K)$ isomorphic zero-dimensional", "metrizable closures π-base", and the source title/DOI. They found the source, Plebanek's π-base theorem, and papers citing the construction, but no later solution of the full question or statement of this localization. The novelty verdict is therefore *candidate new partial reduction*, not a claim of exhaustive bibliographic novelty.

Human-Review Recommendation

Review is recommended. The main checks are narrow: verify that Theorem 3.4 of the source is independent of the choice between ν_1 and ν_2; verify the swapped application of the π-base theorem; and check that an open subset of a metrizable compact closure is F_σ in the ambient compactum. The packet intentionally makes no claim that the zero-set boundary F is metrizable.

References

[1] G. Plebanek, *Musing on Kunen's compact L-space*, Topology Appl. 323 (2023), Paper No. 108294; arXiv:2012.01849.

[2] G. Plebanek, *On isomorphisms of Banach spaces of continuous functions*, Israel J. Math. 209 (2015), 1–13; arXiv:1302.3211.

[3] H. P. Rosenthal, *On injective Banach spaces and the spaces $C(S)$*, Bull. Amer. Math. Soc. 75 (1969), 824–828.